

TEAU 1240 – Automatic Transmissions and Transaxles (4 Credits)

Course Description

Automotive Automatic Transmission and Transaxles is an in-depth clinical based course. Throughout the course, you will practice safety concerns when working on transmissions and drivetrain components, power flow through a planetary gear set, clutch pack diagnosis and operation, hydraulic circuits, removal and rebuilding procedures for automatic transmissions and transaxles, and torque converter operation. When you have completed this course, you will be eligible to take the certification exam in ASE (Automotive Service Excellence) Automotive Automatic Transmissions and Transaxles.

Course Objectives

- Diagnose automatic transmissions and transmission control systems.
- Remove, repair/rebuild and replace automatic transmissions and related systems.
- Diagnose and repair automatic hybrid drive systems.

Course Outline

- Drivetrain Basics
- Drivetrain Theory
- General Theories of Operation
- Electronic Controls
- Transmission Design
- Torque Converters and Pump
- Hydraulic Circuits and Controls
- Gears and Shafts
- Common Transmissions Reaction and Friction Units

Textbook & Reading Materials

Today's Technician: Automatic Transmissions and Transaxles Classroom Manual and Shop Manual, 7th ed. (ISBN: 9781337792158)

Assignments & Assessments

Orientation
Orientation Acknowledgement
Student Information Sheet
Auto Transmissions Course Syllabus Review
Remind Txt Group
Auto Transmissions Competency Profile and Task List Review
Auto Transmissions Lab Assignment Checklist Review
Right-to-Know Agreement
COVID-19 Pandemic Policies and Procedures Agreement
Cleaning Expectations
Southwest Technical College Automotive Video Playlist
Student Tool and Equipment Use Waiver
Cell Phone
Digital Lab Assignment Switch
Instructions
Digital Lab Explanation
Module Breakdown
Module 1 Labs
Module 2 Labs
Module 3 Labs
Module 4 Labs
Module 5 Labs
Module 6 Labs
Module 7 Labs
Module 8 Labs
Module 9 Labs
Cleaning Labs
Cleaning Lab 1
Cleaning Lab 4
Cleaning Lab 3
Cleaning Lab 2
Cleaning Lab 5
Gear Ratios Explained
Planetary Gearset
How a Rear Differential Works
How a Constant Velocity Joint-Axle Works on a FWD
Engine to Rear Axle
Driving Gear and Driven Gear
Freewheeling
Drive Link Chain
Planetary Gear Sets
Chapter Assessments
Chapter 1 Multiple Choice Quiz
Labeling Activity 1-1
Video 1-1 Questions
Lab Assignment 1 Diagnose fluid loss and condition

concerns; check transmission fluid condition; check for leaks.
Lab Assignment 2 Drain and replace fluid and filter(s).
Lab Assignment 3 Check fluid level in a transmission or a transaxle equipped with a dip-stick.
Lab Assignment 4 Inspect power train mounts.
Lab Assignment 5 Identify purpose and demonstrate proper use of fender covers, mats.
Lab Assignment 6 Demonstrate use of the three (concern, cause, and correction).
Lab Assignment 7 Review vehicle history.
Lab Assignment 8 Ensure vehicle is prepared to return to customer.
Checkpoint Meeting Module 1
End Module 1
Reading
Force on Fluid and Measured by the Pressure Gauge
Displaced Area
Hydraulic Pressure and Force Transfer
Chapter 2 Multiple Choice Quiz
Chapter 2 Fill in the Blank Questions
Chapter 2 Short Answer Essay
Labeling Activity 2-1
Chapter 2 ASE-Style Review Questions
Voltage Drop Testing
Diagnostic Process
Repair Orders
Chapter 2 ASE-Style Review Questions
Video 2-1 Questions
Video 2-2 Questions
Lab Assignment 9 Identify and interpret transmission/transaxle concern
Lab Assignment 10 Diagnose fluid loss and condition concerns; check transmission fluid condition; check for leaks.
Lab Assignment 11 Inspect, leak test, and flush or replace transmission/transaxle oil cooler.
Lab Assignment 12 Check fluid level in a transmission or a transaxle not equipped with a dip-stick.
Lab Assignment 13 Check fluid level in a transmission or a transaxle equipped with a dip-stick.
Lab Assignment 14 Research applicable vehicle and service information.
Lab Assignment 15 Identify transmission make and model information.
Lab Assignment 16 Complete work order.

Lab Assignment 17 Check fluid level in a transmission or a transaxle equipped with a dip-stick.

Lab Assignment 18 Demonstrate proper use of precision measuring tools.

Checkpoint Meeting Module 2

End Module 2

Reading

Checking Fluid in a Transmission or Transaxle

Auxiliary Cooler with Inlet and Outlet Tubes

Accumulator Piston

Dual Clutch Transmissions

Operation of a Torque Converter

Transmission Control Module

Chapter 3 Multiple Choice Quiz

Chapter 3 Fill in the Blank Questions

Video 3-1 Questions

Video 3-2 Questions

Reading

Leaks in Front Pump, Front Pump Gasket and Converter

Change Automatic Transmission Filter

Video 3-3 Questions

Chapter 3 ASE-Style Review Questions

Chapter 3 ASE Challenge Questions

Lab Assignment 19 Identify and interpret transmission/transaxle concern.

Lab Assignment 20 Servicing automatic transmission/transaxle linkages.

Lab Assignment 21 Drain and replace fluid and filter(s).

Lab Assignment 22 Diagnose noise and vibration concerns.

Lab Assignment 23 Inspect, leak test, and flush or replace transmission/transaxle oil cooler.

Lab Assignment 24 Remove and reinstall transmission/transaxle and torque converter.

Lab Assignment 25: Check fluid level in a transmission or a transaxle equipped with a dip-stick.

Lab Assignment 26 Inspect converter flex (drive) plate, converter attaching bolts, etc.

Lab Assignment 27 Inspect for leakage at external seals, gaskets, and bushings; replace external seals, gaskets, and bushings.

Lab Assignment 28 Diagnose transmission/transaxle gear reduction/multiplication concerns.

Lab Assignment 29 Perform lock-up converter tests.

Checkpoint Meeting Module 3

End Module 3

CM CH 4 Reading

Using Ohm's Law

Series Circuit

Parallel Circuits

Series-Parallel Circuits

Applying Ohm's Law

Using the Voltmeter

Performing a Voltage Drop Test

Using the Ohmmeter

Using the Ammeter

Reading Wiring Diagrams

Testing For Opens

Testing For Shorts

Testing Switches

Testing Relays

Using the Lab Scope

Using the Scan Tool

Checking for Codes and Monitor Status

Electronic Shifting

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Electrons Passing to Protons Video/Animation

Wave on Voltage Measurement Over Time 1 Video/Animation

Wave on Voltage Measurement Over Time 2 Video/Animation

Wave on Voltage Measurement Over Time, Rising and Falling

Slopes Video/Animation

Ohm's Law Video/Animation

Overview of a constant variable transmission Video/Animation

CM CH 4 Video 4-1 Questions

CM CH 4 Video 4-2 Questions

CM CH 4 Multiple Choice Quiz

CM CH 4 Fill in the Blank Questions

SM CH 4 Reading

Horn and Brake Lamps Circuit Video/Animation

Circuit Breaker, Switch and Lamp Video/Animation

Voltage Measurement Over Time Video/Animation

Noise and Glitches Video/Animation

Variable Resistance Crystal Video/Animation

Scan Tools Video/Animation

Purpose of Diagnostic Trouble Code (DTC) Video/Animation

Single Circuit Schematics Video/Animation

Transmission Solenoid Video/Animation

SM CH 4 Video 4-3 Questions

SM CH 4 Video 4-4 Questions

SM CH 4 ASE-Style Review Questions

SM CH 4 ASE Challenge Questions

Lab Assignment 30 Diagnose electrical/electronic control systems.

Lab Assignment 31 Inspect, test, adjust, repair, or replace

electrical/electronic components.

Lab Assignment 32 Inspect, test, adjust, repair, or replace electrical/electronic components.

Lab Assignment 33 Describe the operational characteristics of a continuously variable transmission.

Lab Assignment 34 Describe the operational characteristics of a hybrid vehicle drive train.

Checkpoint Meeting Module 4

End Module 4

CM CH 5 Reading: Transmission Designs

Cylinder Wall with Seal Lip and Piston Video/Animation

Types and Usage of Seals and Gaskets Video/Animation

CM CH 5 Labeling Activity 5-1: Differential Parts ID

CM CH 5 Multiple Choice Quiz: Transmission Designs

SM CH 5 Reading: Rebuilding Transmissions and Transaxles

SM CH 5 Labeling Activity 5-2: External Transaxle Parts ID

SM CH 5 Photo Sequence 8: Measuring Input Shaft Thrust Play (Endplay)

SM CH 5 ASE-Style Review Questions: Rebuilding Transmissions and Transaxles

SM CH 5 ASE Challenge Questions: Rebuilding Transmissions and Transaxles

Lab Assignment 35 Perform pressure tests.

Lab Assignment 36 Remove and reinstall transmission/transaxle and torque converter.

Lab Assignment 37 Measure torque converter end play and check for interference.

Lab Assignment 38 Perform stall test.

Lab Assignment 39 Diagnose pressure concerns.

Checkpoint Meeting Module 5

End Module 5

Reading

Torque Converter Clutch Operation

Torque Converter Clutch Operation to Drive Input Shaft Converter at Coupling Speed, Stator Overrunning Video/Animation

Torque Converter Video/Animation

CM CH 6 Video 6-1 Questions: Operation of a torque converter

CM CH 6 Labeling Activity 6-1: Torque Converter Parts ID

CM CH 6 Multiple Choice Quiz: Torque Converters and Pumps

SM CH 6 Reading: Torque Converter and Oil Pump Service

Gear with Gauge 1 Video/Animation

Gear with Gauge 2 Video/Animation

Operation of a Torque Converter Lockup Clutch (TCC) Video/Animation

SM CH 6 Labeling Activity 6-2: Auto Transmission Pump

Parts ID

SM CH 6 Video 6-2 Questions: Operation of a torque converter lock-up clutch

SM CH 6 ASE-Style Review Questions: Torque Converter and

Oil Pump Service

SM CH 6 ASE Challenge Questions: Torque Converter and Oil

Pump Service

Lab Assignment 40 Disassemble, measure, inspect, and reassemble an automatic transmission.(Tasks

C6,7,8,9,10,11,12,13,14,15,16,19,20,21,22)

Checkpoint Meeting Module 6

End Module 6

CM CH 7 Reading

Fluid Flow Video/Animation

Fluid Flow with Spring Pressure Video/Animation

Function, Construction, and Operation of a Spool Valve Video/Animation

Types and Functions of Automatic Transmission Valves Video/Animation

Types of Spool Valves Video/Animation

CM CH 7 Video 7-1 Questions

CM CH 7 Video 7-2 Questions

CM CH 7 Multiple Choice Quiz

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SM CH 7 Video 7-3 Questions

SM CH 7 ASE-Style Review Questions

SM CH 7 ASE Challenge Questions

Lab Assignment 41 Disassemble, measure, inspect, and reassemble an automatic transmission.(Tasks

C6,7,8,9,10,11,12,13,14,15,16,19,20,21,22)

Checkpoint Meeting Module 7

End Module 7

CM CH 8 Reading

Walk Around the Sun Gear Video/Animation

Walk Inside Ring Gear Video/Animation

Gear ratios Video/Animation

CM CH 8 Labeling Activity 8-1

CM CH 8 Video 8-1 Questions

CM CH 8 Multiple Choice Quiz

SM CH 8 Reading

SM CH 8 Labeling Activity 8-2

SM CH 8 Photo Sequence 13: Servicing Planetary Gear-Type

Final Drive Units

SM CH 8 ASE-Style Review Questions

SM CH 8 ASE Challenge Questions

transmission.(Tasks
C6,7,8,9,10,11,12,13,14,15,16,19,20,21,22)
Checkpoint Meeting Module 8
End Module 8
CM CH 9 Reading
Locked and Unlocked Gears Video/Animation
Automatic Transmission Rear Servo Operation
Video/Animation
Multi-disc Hydraulic Clutch Video/Animation
CM CH 9 Labeling Activity 9-1
CM CH 9 Video 9-1 Questions
CM CH 9 Multiple Choice Quiz
CM CH 9 Fill in the Blank Questions
SM CH 9 Reading
Service Limit Video/Animation
SM CH 9 Photo Sequence 14: Proper Procedure for
Installing
a Direct Clutch
SM CH 9 Labeling Activity 9-2
SM CH 9 ASE-Style Review Questions

SM CH 9 ASE Challenge Questions
CM CH 10 Reading
How a Dual Clutch Transmission Operates Simulation
CM CH 10 Labeling Activity 10-1
CM CH 10 Multiple Choice Quiz
SM CH 10 Reading
SM CH 10 Labeling Activity 10-2
SM CH 10 Labeling Activity 10-3
SM CH 10 ASE-Style Review Questions
SM CH 10 ASE Challenge Questions
Lab Assignment 43 Disassemble, measure, inspect, and
reassemble an automatic transmission.(Tasks
C6,7,8,9,10,11,12,13,14,15,16,19,20,21,22)
Checkpoint Meeting Module 9
End Module 9
End of Course Survey
Competency Profile
Final Exam Review
Final Exam

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Mo, Tu, W, Th

8:00 AM - 12:00 AM

1:00 PM - 5:00 PM

Friday

8:00 AM - 12:00 AM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

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Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

Course Grading: Students must achieve 80% (B-) or higher to pass graded work. Incomplete assignments must be redone to meet the required standards. Guidelines, rules, and expectations for completing assignments are provided in each course.

High School Power School Grades: Quarter student grades will be determined by student progress percentage. Faculty will use the higher percentage of either 1) quarter progress, or 2) cumulative progress for the current training plan year.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

Class attendance is required during your scheduled time. Cell phones for many have become a distraction. When you are in class or lab we encourage you to keep your cell phones put away in a secure location. If you use ear buds we ask that you only use one so you can still hear the things going on around you. If you are using your phone for things other than school related items, instructors will ask you to put them away. Please follow the direction of your instructors. Those who have been asked to refrain from using your cell phone and fail to do so will be asked to meet with the Director of Transportation and student services will be notified. The program is designed to provide the student with as much hands-on work as possible. In the automotive industry you may be required to lift heavy objects and stand for hours at a time to complete work required. Technicians deal with chemicals and materials which require caution, these will be identified in the Right to Know Agreement provided to you. You will also be required to use computers to track and complete work.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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